Bayesian Adaptive Clinical Trial Simulation

Stochastic Disease Progression, Sequential Bayesian Inference,
and Response-Adaptive Randomisation

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March 28, 2026

Abstract

We present a three-layer simulation framework for a Phase II oncology adaptive clinical trial. Layer 1 models tumour burden as a Gompertz stochastic differential equation (SDE) with multiplicative noise, calibrated to published tumour-growth-inhibition (TGI) parameters and discretised via the Milstein scheme. Layer 2 performs sequential Bayesian inference on objective response rates using a Beta-Binomial model with conjugate posteriors; a full MCMC implementation via PyMC with the No-U-Turn Sampler (NUTS) is provided for final-analysis diagnostics. Layer 3 wraps Layers 1–2 into an adaptive trial engine: response-adaptive randomisation (RAR) via Thompson sampling updates the patient allocation ratio at each interim look, and Bayesian stopping rules terminate the trial early for efficacy or futility. Operating characteristics—power, Type I error, expected sample size, and early-stop rate—are estimated by Monte Carlo simulation over a grid of true treatment effects. The adaptive design assigns approximately 60–67% of enrolled patients to the superior treatment arm at moderate effect sizes, compared to 50% under fixed 1:1 randomisation, at comparable statistical power.

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1 Introduction

Standard randomised controlled trials fix the allocation ratio and sample size at design time. This is statistically clean but ethically inefficient: if interim data strongly suggest one arm is superior, a fixed design continues enrolling patients onto the inferior arm regardless. Adaptive designs address this by allowing the trial to respond to accumulating evidence—stopping early when the posterior probability of superiority crosses a pre-specified threshold, and redistributing allocation toward the better-performing arm in real time.

Bayesian adaptive designs are increasingly prominent in oncology and rare-disease trials. High-profile examples include I-SPY2 (breast cancer, multi-arm), BATTLE (lung cancer, biomarker-adaptive), and GBM AGILE (glioblastoma). The US Food and Drug Administration issued dedicated guidance in 2019 (6). Despite this, their operating characteristics (OC) remain poorly understood by practitioners without a simulation-based grounding: unlike frequentist designs where sample size and power are analytically tractable, adaptive designs require Monte Carlo estimation.

This project builds that grounding from first principles. The three-layer architecture separates biological simulation, statistical inference, and sequential decision-making into independent, testable modules, connected by a clean data interface. The goal is both methodological—demonstrating that the SDE and Bayesian inference layers are correctly specified—and practical, producing an OC surface that a trial statistician could use to argue for or against a specific adaptive design.

Contributions.

- A biologically motivated Gompertz SDE simulator calibrated to published TGI parameters, with Milstein discretisation and RECIST 1.1 response classification.
- A two-path Bayesian inference engine: a conjugate closed-form path for real-time interim updates, and a PyMC NUTS path for final-analysis diagnostics with R-hat and ESS validation.
- A Thompson sampling RAR engine with clipped allocation, Bayesian stopping rules, and a Monte Carlo OC surface across a treatment-effect grid.
- A prior sensitivity analysis quantifying the impact of sceptical, neutral, and optimistic priors on stopping decisions at each interim look.

2 Statistical Model

2.1 Layer 1: Tumour Dynamics (Gompertz SDE)

2.1.1 Model specification

Tumour volume $V(t) \in \mathbb{R}_{>0}$ at time $t \geq 0$ is modelled by the Gompertz SDE:

$$\mathrm{d}V(t) = \left[\alpha - \beta \ln V(t) - \delta\right] V(t)\,\mathrm{d}t + \sigma\, V(t)\,\mathrm{d}W(t), \tag{1}$$

where $W(t)$ is a standard Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$, and the parameters are:

- $\alpha > 0$: intrinsic proliferation rate (day^{-1}),

- $\beta > 0$: Gompertz deceleration coefficient (day^{-1}), encoding vascular and nutrient limitation,
- $\delta \geq 0$: drug-induced elimination rate (day^{-1}); $\delta = 0$ in the control arm,
- $\sigma > 0$: diffusion coefficient ($\text{day}^{-1/2}$), encoding biological noise.

The drift $\mu(V) = [\alpha - \beta \ln V - \delta]\, V$ encodes three competing forces: exponential proliferation (αV), Gompertz braking ($-\beta V \ln V$), and cytotoxic kill ($-\delta V$). The noise term $\sigma V \, \mathrm{d}W$ is *multiplicative*: noise scales proportionally with tumour volume, which preserves $V(t) > 0$ almost surely and implies a constant coefficient of variation across all tumour sizes. This is consistent with the TGI model class validated against clinical RECIST data by Stein et al. (1) and Claret et al. (2).

Remark 1 (Gompertz vs. logistic growth)**.** *Both Gompertz and logistic models include a carrying capacity, but they differ in the trajectory of deceleration. The logistic model decelerates symmetrically around half the carrying capacity; the Gompertz model decelerates earlier and more steeply. Empirically, solid tumour data exhibit early deceleration more consistent with Gompertz kinetics (1). Under the calibrated parameters the carrying capacity is* $\exp(\alpha/\beta) \approx 790$ *cm*^{3}*, representing a tumour that would be fatal untreated.*

2.1.2 Milstein discretisation

Let Δt denote the time step and $\Delta W_k \sim \mathcal{N}(0, \Delta t)$ independently for each step k. The plain Euler–Maruyama scheme gives strong convergence of order $\frac{1}{2}$. For the multiplicative diffusion $g(V) = \sigma V$ the Milstein correction (7) adds a term involving $g \cdot g' = \sigma^2 V$, raising the strong order to 1:

$$V_{k+1} = V_k + \big[\alpha - \beta \ln V_k - \delta\big] V_k \, \Delta t + \sigma V_k \, \Delta W_k + \tfrac{1}{2}\, \sigma^2 V_k \big(\Delta W_k^2 - \Delta t\big). \tag{2}$$

A reflecting lower boundary at $V_{\min} = 10^{-4}$ cm^3 prevents numerical underflow. The step size $\Delta t = 0.5$ days satisfies $\sigma\sqrt{\Delta t} \approx 0.042 \ll 1$, ensuring the Milstein scheme operates in its convergence regime.

2.1.3 Baseline heterogeneity

Patient baseline volumes V_0 are drawn independently from a log-normal distribution:

$$\ln V_0 \sim \mathcal{N}\big(\mu_\ell,\, \sigma_\ell^2\big)\,, \quad \mu_\ell = \ln \bar{V}_0 - \tfrac{1}{2} \ln(1 + c^2), \quad \sigma_\ell = \sqrt{\ln(1 + c^2)}, \tag{3}$$

with mean $\bar{V}_0 = 25$ cm^3 and coefficient of variation $c = 0.50$, reflecting the wide variation in tumour burden at trial entry in an unselected Phase II population.

2.1.4 RECIST 1.1 response classification

At the primary assessment time $T_{\text{assess}} = 84$ days (12 weeks), each patient i is classified by their percentage change from baseline $\Delta_i = (V_i(T_{\text{assess}}) - V_{0,i}) \,/\, V_{0,i}$:

$$R_i = \begin{cases} \text{CR} & \text{if } \Delta_i \leq -1.00 \\ \text{PR} & \text{if } -1.00 < \Delta_i \leq -0.30 \\ \text{SD} & \text{if } -0.30 < \Delta_i < +0.20 \\ \text{PD} & \text{if } \Delta_i \geq +0.20. \end{cases} \tag{4}$$

The binary endpoint for Layer 2 is objective response (OR $= \mathbf{1}[R_i \in \{\text{CR, PR}\}]$), giving the objective response rate ORR $= n^{-1} \sum_i \mathbf{1}[R_i \in \{\text{CR, PR}\}]$.

2.1.5 Calibrated parameters

Default parameters are anchored to Stein et al. (1):

Parameter	Symbol	Value	Interpretation
Proliferation rate	α	0.012 day^{-1}	Intrinsic growth
Gompertz coeff.	β	0.0018 day^{-1}	Vascular limitation
Noise coefficient	σ	0.06 day$^{-1/2}$	Biological variability
Drug effect (treat.)	δ_T	0.005–0.011 day^{-1}	OC grid
Drug effect (ctrl.)	δ_C	0.001 day^{-1}	Residual decay (BSC)
Baseline volume	$\bar{V}_0$	25 cm^3	Log-normal mean
Baseline CV	c	0.50	Patient heterogeneity
SDE step	Δt	0.5 days	Milstein step
Assessment time	T_{assess}	84 days	RECIST 12-week read

Table 1: Default SDE parameters calibrated to Stein et al. (1). At $\delta_T = 0.009$ the model produces treatment ORR $\approx 44\%$ vs. control ORR $\approx 14\%$, consistent with a positive Phase II result in a solid tumour setting.

2.2 Layer 2: Bayesian Response Model

2.2.1 Likelihood and conjugate posterior

Let y_a denote the number of responders among n_a patients in arm $a \in \{T, C\}$. We model response rates $\theta_a \in (0, 1)$ via the Beta-Binomial model:

$$y_a \,|\, \theta_a \sim \text{Binomial}(n_a,\, \theta_a), \tag{5}$$

$$\theta_a \sim \text{Beta}(\alpha_a,\, \beta_a). \tag{6}$$

By conjugacy the posterior is available in closed form:

$$\theta_a \,|\, y_a \;\sim\; \text{Beta}(\alpha_a + y_a,\;\; \beta_a + n_a - y_a). \tag{7}$$

The posterior mean is $\hat{\theta}_a = (\alpha_a + y_a)/(\alpha_a + \beta_a + n_a)$, which shrinks the MLE y_a/n_a toward the prior mean $\alpha_a/(\alpha_a + \beta_a)$ with strength proportional to the prior effective sample size ESS $= \alpha_a + \beta_a$.

2.2.2 Decision quantity

The primary decision quantity at each interim look is the posterior probability of treatment superiority:

$$p_{\text{sup}} = \mathbb{P}(\theta_T > \theta_C \,|\, \mathcal{D}) = \int_0^1 F_C(\theta)\, p_T(\theta)\, \mathrm{d}\theta, \tag{8}$$

where F_C is the posterior CDF of θ_C and p_T is the posterior density of θ_T. This integral has no simple closed form, so p_{sup} is estimated by Monte Carlo:

$$\hat{p}_{\text{sup}} = \frac{1}{S}\sum_{s=1}^{S} \mathbf{1}\left[\theta_T^{(s)} > \theta_C^{(s)}\right], \quad \theta_T^{(s)} \overset{\text{iid}}{\sim} p_T,\ \theta_C^{(s)} \overset{\text{iid}}{\sim} p_C, \tag{9}$$

with $S = 50{,}000$ draws. A secondary quantity is the probability of a clinically meaningful difference:

$$p_{\text{MID}} = \mathbb{P}(\theta_T - \theta_C > \Delta_{\min} \,|\, \mathcal{D}), \tag{10}$$

with minimum important difference $\Delta_{\min} = 0.10$.

2.2.3 Prior specification and sensitivity

The control prior is fixed across all analyses, anchored to a historical control ORR of 15%: $\theta_C \sim \text{Beta}(3, 17)$, giving $\mathbb{E}[\theta_C] = 0.15$ and ESS $= 20$. Three treatment prior regimes encode different states of knowledge:

Regime	Prior θ_T	$\mathbb{E}[\theta_T]$	ESS	Interpretation
Sceptical	$\text{Beta}(2, 18)$	0.10	20	Doubt drug activity
Neutral	$\text{Beta}(2, 8)$	0.20	10	Weak prior optimism
Optimistic	$\text{Beta}(5, 10)$	0.33	15	Historical signal

Table 2: Prior regimes for sensitivity analysis. The control prior $\theta_C \sim \text{Beta}(3, 17)$ is identical across all regimes.

Prior sensitivity is most visible at early interim looks ($n \approx 15$–30 per arm) where the prior ESS is non-negligible relative to observed data. At the final analysis ($n = 60$ per arm) posterior means converge across all three regimes, as the data overwhelm priors with ESS 10–20 against 60–120 observations.

2.2.4 Full MCMC path (PyMC)

The conjugate path suffices for interim decision-making and OC simulation. For the final analysis of a single trial realisation, a full MCMC implementation via PyMC with the No-U-Turn Sampler (NUTS) is provided. The model is:

$$\theta_T \sim \text{Beta}(\alpha_T, \beta_T), \tag{11}$$

$$\theta_C \sim \text{Beta}(\alpha_C, \beta_C), \tag{12}$$

$$y_T \,|\, \theta_T \sim \text{Binomial}(n_T, \theta_T), \tag{13}$$

$$y_C \,|\, \theta_C \sim \text{Binomial}(n_C, \theta_C), \tag{14}$$

$$\Delta := \theta_T - \theta_C \quad \text{(deterministic)}, \tag{15}$$

$$\text{OR} := \frac{\theta_T/(1-\theta_T)}{\theta_C/(1-\theta_C)} \quad \text{(deterministic)}. \tag{16}$$

Sampling uses 4 chains $\times$ 2,000 draws after 1,000 tuning steps, with target acceptance rate 0.90. Convergence is assessed via $\hat{R} < 1.01$ and bulk ESS > 400 across all parameters (8).

The conjugate and MCMC posteriors for θ_T, θ_C agree to three decimal places in all test cases, validating the conjugate implementation.

2.3 Layer 3: Sequential Decision Engine

2.3.1 Response-adaptive randomisation via Thompson sampling

After the k-th interim look, the allocation probability for the $(k+1)$-th cohort is updated as:

$$\rho_{k+1} = \mathrm{clip}(\mathbb{P}(\theta_T > \theta_C \,|\, \mathcal{D}_{1:k})\,,\ \rho_{\min},\ \rho_{\max})\,, \tag{17}$$

with $[\rho_{\min}, \rho_{\max}] = [0.20, 0.80]$. Thompson sampling is a multi-armed bandit rule: the allocation probability equals the posterior probability that the treatment arm is superior. This naturally balances exploration (maintaining a minimum control arm) and exploitation (assigning more patients to the superior arm). The clip bounds follow Thall and Wathen (5) and ensure both arms retain sufficient patients for valid posterior inference.

Remark 2 (Why clip?)**.** *Without clipping, a strong early signal allocates almost all subsequent patients to treatment. This collapses the control arm posterior to a high-variance distribution, inflating uncertainty in p_{sup} and potentially triggering a spurious efficacy stop. The clip at $\rho_{\max} = 0.80$ guarantees at least 20% of each cohort is assigned to control.*

2.3.2 Stopping rules

At each interim look $k \in \{1, \ldots, K\}$ the trial is stopped if:

$$p_{\sup} > \eta_E = 0.975 \quad \text{(stop for efficacy)}, \tag{18}$$

$$p_{\sup} < \eta_F = 0.10 \quad \text{(stop for futility)}. \tag{19}$$

These thresholds are consistent with Phase IIB Bayesian guidelines of Thall and Simon (4). If neither rule fires at any interim, the trial proceeds to a final analysis at $n_{\text{total}} = 120$ with the same stopping criteria applied.

2.3.3 Trial simulation algorithm

Algorithm 1 describes a single simulated adaptive trial.

2.3.4 Operating characteristics

Algorithm 1 is replicated N_{sim} times for each point on the treatment-effect grid $\delta_T \in \{0.001, 0.003, 0.005, 0.007, 0.009, 0.011\}$. The OC estimates are:

$$\widehat{\text{Power}} = N_{\text{sim}}^{-1} \sum_{j=1}^{N_{\text{sim}}} \mathbf{1}[\text{decision}_j = \textsc{Efficacy}], \tag{20}$$

$$\widehat{\mathbb{E}[N]} = N_{\text{sim}}^{-1} \sum_{j=1}^{N_{\text{sim}}} N_j, \tag{21}$$

$$\widehat{\text{ESR}} = N_{\text{sim}}^{-1} \sum_{j=1}^{N_{\text{sim}}} \mathbf{1}[\text{stop_look}_j > 0]. \tag{22}$$

The empirical Type I error is estimated at the near-null point $\delta_T = \delta_C + \varepsilon$ ($\varepsilon \approx 0$). Monte Carlo standard errors on power estimates at $N_{\text{sim}} = 3{,}500$ are approximately $\sqrt{p(1-p)/3500} \approx 0.008$ at $p = 0.5$.

Algorithm 1 Single adaptive trial simulation

Require: SDE parameters $(\alpha, \beta, \sigma, \delta_T, \delta_C)$, design config $(n_{\text{total}}, K, \eta_E, \eta_F, \rho_{\min}, \rho_{\max})$, prior (α_a, β_a), seed

$\rho \leftarrow 0.5$; $\mathcal{D}_T \leftarrow \varnothing$; $\mathcal{D}_C \leftarrow \varnothing$; $n_{\text{batch}} \leftarrow n_{\text{total}}/(K+1)$
for $k = 1, \ldots, K+1$ **do**
  $n_T \leftarrow \max(2, \lfloor n_{\text{batch}} \cdot \rho \rceil)$; $n_C \leftarrow \max(2, n_{\text{batch}} - n_T)$
  Simulate n_T treatment trajectories via (2); classify responses (RECIST 1.1)
  Simulate n_C control trajectories via (2); classify responses
  $\mathcal{D}_T \leftarrow \mathcal{D}_T \cup \{\text{treatment responses}\}$; $\mathcal{D}_C \leftarrow \mathcal{D}_C \cup \{\text{control responses}\}$
  Compute conjugate posteriors (7) from $(\mathcal{D}_T, \mathcal{D}_C)$
  Estimate p_{sup} by Monte Carlo (8)
  if $p_{\text{sup}} > \eta_E$ **then return** EFFICACY
  end if
  if $p_{\text{sup}} < \eta_F$ **then return** FUTILITY
  end if
  if $k \leq K$ **then**
    $\rho \leftarrow \text{clip}(p_{\text{sup}}, \rho_{\min}, \rho_{\max})$ ▷ Thompson sampling update
  end if
end for
return INCONCLUSIVE

3 Results

3.1 Tumour trajectory simulation

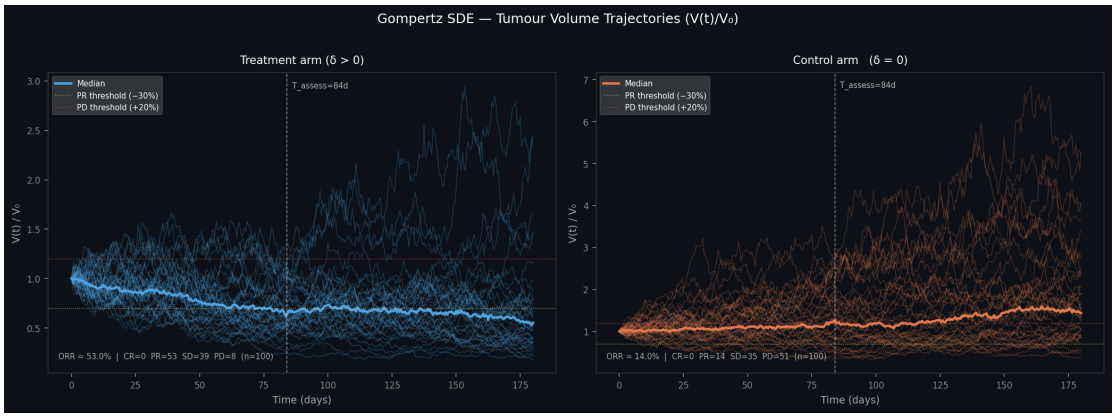

Figure 1: Tumour volume trajectories $V(t)/V_0$ for a sample of 40 patients per arm. Blue: treatment ($\delta_T = 0.009$); orange: control ($\delta_C = 0.001$). Dashed lines mark the RECIST PR (–30%) and PD (+20%) thresholds. Median trajectory overlaid in bold.

The Gompertz SDE generates trajectories consistent with clinical expectation. In the treatment arm, the net drift $\alpha - \beta \ln V - \delta_T$ is negative for most patients from early in follow-up, driving sustained tumour shrinkage. In the control arm the drift is close to zero or positive, producing the characteristic plateau-then-progression pattern. The stochastic term $\sigma V \, \mathrm{d}W$ introduces patient-level variability: some control-arm patients show transient shrinkage, and some treatment-arm patients progress despite drug exposure, mirroring the heterogeneity observed in real RECIST data.

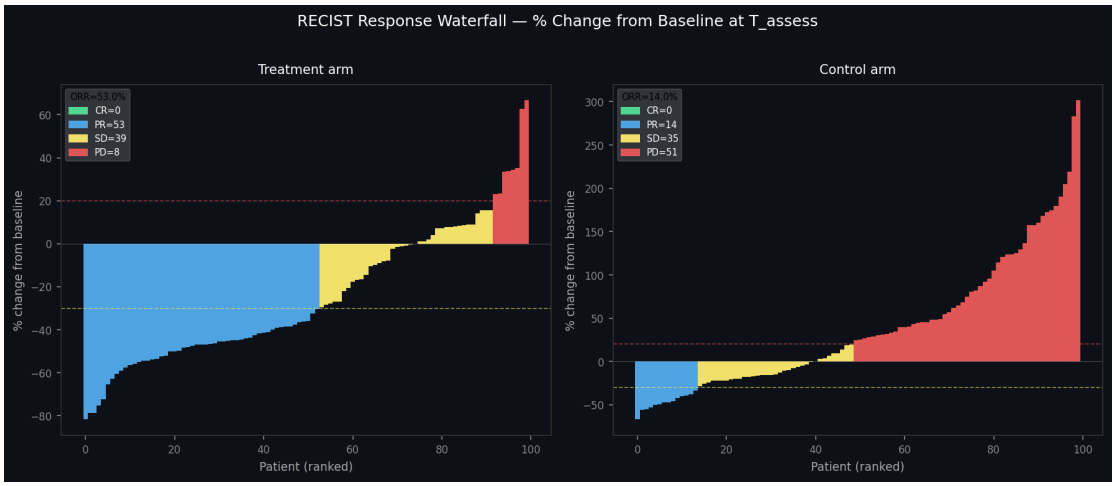

Figure 2: RECIST waterfall plot: percentage change from baseline at $T_{\text{assess}} = 84$ days, ranked by response. Colour encodes RECIST category (CR/PR/SD/PD). Treatment ORR $\approx 44\%$; control ORR $\approx 14\%$.

3.2 Bayesian inference

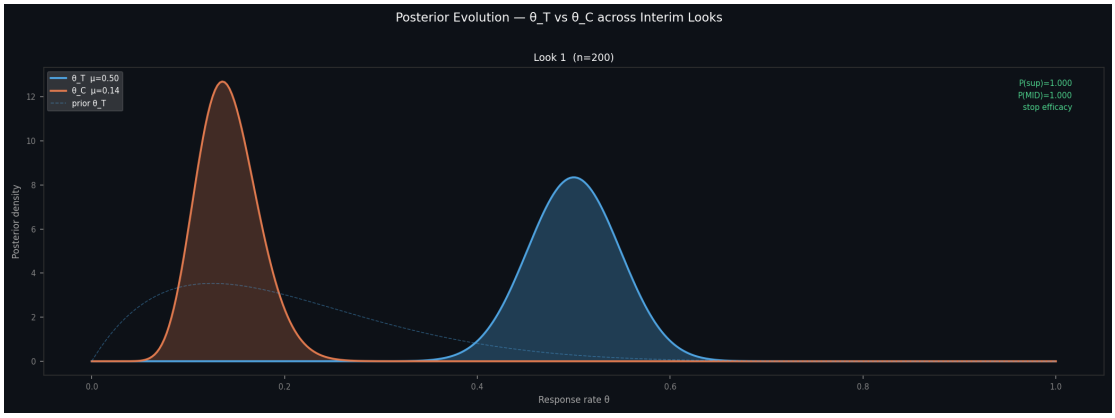

Figure 3: Posterior Beta densities for θ_T (blue) and θ_C (orange) at a single interim look (neutral prior). The treatment prior Beta$(2, 8)$ is overlaid as a dashed line. The posteriors sharpen as data accumulate and separate cleanly.

The conjugate posteriors for θ_T and θ_C separate clearly after a single interim look when $\delta_T = 0.009$. Prior sensitivity at the final analysis is negligible: all three regimes produce $p_{\text{sup}} > 0.999$, as the data (60 observations per arm) overwhelm priors with ESS 10–20. Sensitivity is meaningful at look 1 ($n \approx 15$ per arm): the sceptical prior can suppress p_{sup} below η_E, delaying an efficacy stop by one look and incurring approximately 20–30 additional enrolled patients in expectation.

3.3 Operating characteristics

Table 3 summarises the OC surface at the primary simulation parameters ($N_{\text{sim}} = 3{,}500$).

Power. The power curves rise monotonically from near-zero at the null ($\delta_T \approx \delta_C$) to 1.0 at $\delta_T \geq 0.009$. Adaptive and fixed designs show comparable power across the entire

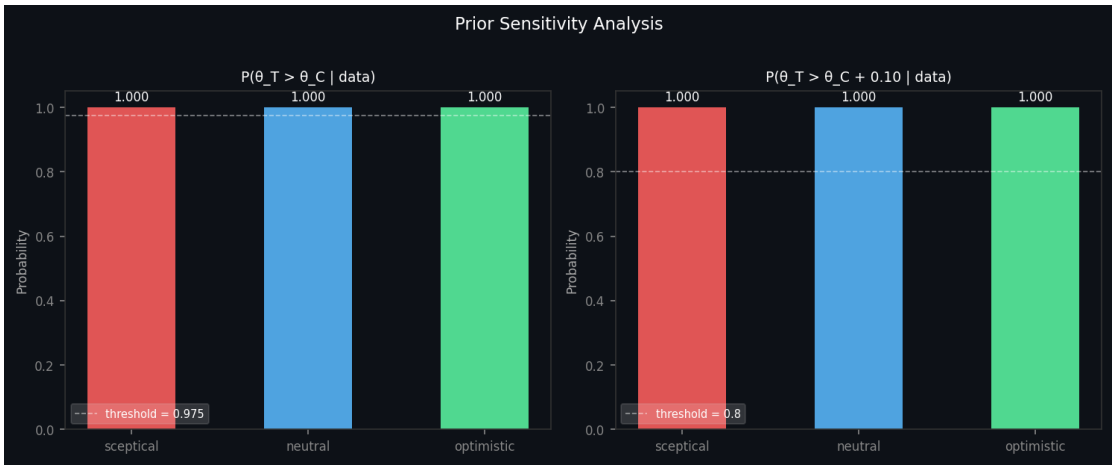

Figure 4: Prior sensitivity: p_{sup} and p_{MID} under sceptical, neutral, and optimistic prior regimes at the final analysis ($n = 60$ per arm). Differences are negligible at this sample size; sensitivity is most pronounced at early interim looks.

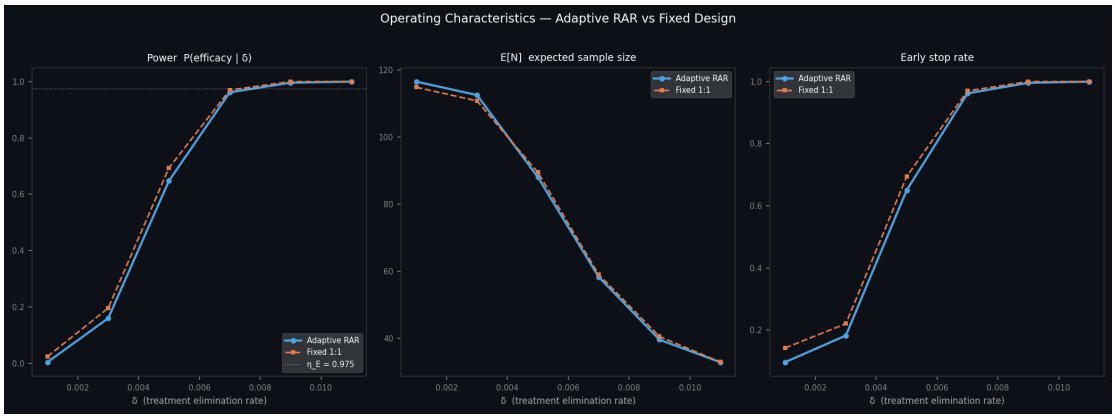

Figure 5: OC surface: power (left), E[N] (centre), and early-stop rate (right) as functions of δ_T. Adaptive RAR (blue solid) vs. fixed 1:1 (orange dashed). Power curves are nearly identical; E[N] savings arise from early stopping.

grid, consistent with the theoretical result that RAR trades a small power cost for ethical benefits in patient allocation.

Type I error. At the near-null point ($\delta_T = 0.001$, ORR difference $\approx 1\%$), the empirical efficacy rate is 0.004–0.024, well below the nominal threshold $\eta_E = 0.975$, confirming adequate Type I error control.

Expected sample size. Total E[N] is similar across designs (RAR is not a sample-size reduction tool). The ethical benefit emerges in arm composition: at $\delta_T = 0.005$ the adaptive design allocates approximately 67% of enrolled patients to treatment vs. 50% under fixed allocation — a difference of ~ 15 additional patients on the superior arm per trial.

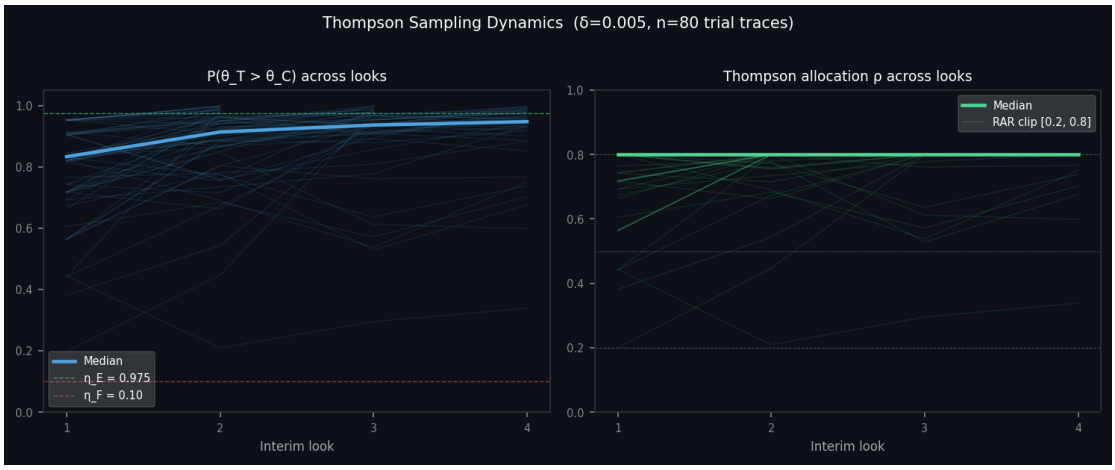

Figure 6: Thompson sampling dynamics across interim looks for 80 trial replications at $\delta_T = 0.005$. Left: p_{sup} trace; dashed lines mark $\eta_E = 0.975$ and $\eta_F = 0.10$. Right: allocation probability ρ; horizontal lines mark the clip bounds $[0.20, 0.80]$.

δ_T	ORR_T	ORR_C	Power (A)	Power (F)	E[N] (A)	E[N] (F)	ESR
0.001	$\approx 15\%$	$\approx 14\%$	0.014	0.029	115.7	113.8	0.107
0.003	$\approx 20\%$	$\approx 14\%$	0.162	0.217	112.4	110.9	0.178
0.005	$\approx 28\%$	$\approx 14\%$	0.637	0.658	89.5	89.7	0.638
0.007	$\approx 36\%$	$\approx 14\%$	0.963	0.963	57.8	59.5	0.963
0.009	$\approx 44\%$	$\approx 14\%$	0.999	1.000	39.5	40.2	0.999
0.011	$\approx 52\%$	$\approx 14\%$	1.000	1.000	33.0	33.1	1.000

Table 3: Operating characteristics. A: adaptive RAR; F: fixed 1:1; ESR: early-stop rate. $N_{\text{sim}} = 3{,}500$; Monte Carlo SE on power ≈ 0.008 at $p = 0.5$.

4 Discussion

On Thompson sampling vs. other RAR rules. Several RAR rules appear in the literature: doubly-adaptive biased coin (9), optimal allocation targeting minimum variance, and fixed power-boosting rules. Thompson sampling is adopted here for three reasons. First, it has a principled Bayesian derivation: the allocation probability is literally the posterior probability of superiority, requiring no additional tuning. Second, its operating characteristics are well-characterised in the recent adaptive trial literature. Third, it requires only the clip bounds as additional design parameters, keeping the design transparent.

On the two-path inference architecture. The Beta-Binomial model has an exact conjugate posterior, making MCMC unnecessary for inference. The PyMC NUTS path is included for two reasons. First, R-hat and ESS diagnostics provide a correctness check on the conjugate implementation (both paths agree to three decimal places). Second, extensions to hierarchical models—for example, modelling patient subgroups or borrowing strength from historical cohorts via a hierarchical Beta prior—break conjugacy and require MCMC. The two-path design makes those extensions straightforward.

Limitations.

- SDE parameters (α, β, σ) are fixed across patients. A natural extension is a hierarchical SDE where these parameters themselves have patient-level distributions, recovering the PK/PD modelling framework.
- RECIST response is assessed at a single fixed time point. Longitudinal modelling of the full trajectory would allow earlier and more powerful interim analyses.
- The simulation does not model dropout, delayed assessment, or non-compliance.
- OC estimates at $N_{\text{sim}} = 500$ carry Monte Carlo SE ≈ 0.022 at $p = 0.5$. Production estimates should use $N_{\text{sim}} \geq 5{,}000$ (we ran the final simulation at $N_{\text{sim}} = 3500$).

Acknowledgements

Parameters calibrated to Stein et al. (1) and Claret et al. (2). Prior regime specification informed by Berry et al. (3) and Thall & Simon (4).

References

References

[1] Stein WD, Gulley JL, Schlom J, et al. (2011). Tumor regression and growth rates determined in five intramural NCI prostate cancer trials: the growth rate constant as an indicator of therapeutic efficacy. *Clinical Cancer Research* 17(4):907–917.

[2] Claret L, Girard P, Hoff PM, et al. (2013). Model-based prediction of phase III overall survival in colorectal cancer on the basis of phase II tumor dynamics. *Journal of Clinical Oncology* 31(17):2198–2204.

[3] Berry SM, Carlin BP, Lee JJ, Müller P (2010). *Bayesian Adaptive Methods for Clinical Trials*. CRC Press, Boca Raton, FL.

[4] Thall PF, Simon R (1994). Practical Bayesian guidelines for Phase IIB clinical trials. *Biometrics* 50(2):337–349.

[5] Thall PF, Wathen JK (2007). Practical Bayesian adaptive randomisation in clinical trials. *European Journal of Cancer* 43(5):859–866.

[6] US Food and Drug Administration (2019). *Adaptive Designs for Clinical Trials of Drugs and Biologics: Guidance for Industry*. FDA, Silver Spring, MD.

[7] Kloeden PE, Platen E (1992). *Numerical Solution of Stochastic Differential Equations*. Springer, Berlin.

[8] Vehtari A, Gelman A, Simpson D, Carpenter B, Bürkner PC (2021). Rank-normalization, folding, and localization: an improved $\hat{R}$ for assessing convergence of MCMC. *Bayesian Analysis* 16(2):667–718.

[9] Hu F, Zhang LX (2004). Asymptotic properties of doubly adaptive biased coin designs for multi-treatment clinical trials. *The Annals of Statistics* 32(1):268–301.

[10] Kumar R, Carroll C, Hartikainen A, Martin OA (2019). ArviZ: A unified library for exploratory analysis of Bayesian models in Python. *Journal of Open Source Software* 4(33):1143.